NFS (Network File System) Configuration

Cluster: Virtual HPC Cluster

NFS Server: storage (192.168.100.12)

NFS Clients: hpc-master, compute1, compute2, login

Shared Directories: /home, /soft, /opt

1. What is NFS?

NFS (Network File System) is a **distributed file system protocol** that allows a server to **share directories over a network**, and clients to mount and access those directories as if they were local storage.

In an HPC cluster, NFS is essential because:

- **Users log in from any node** — their home directories must be available everywhere
- **Software is installed once** — applications in /opt or /soft are shared across all nodes
- **Jobs run on compute nodes** — input/output files must be accessible from all compute nodes simultaneously
- **No duplication** — instead of copying files to each node, one central copy is shared

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2. NFS Architecture in This Cluster

In this cluster, the **storage node is dedicated as the NFS server**. It exports three shared directories to all other nodes over the cluster's internal network (192.168.100.0/24).

```
storage (NFS Server — 192.168.100.12)
        │
        │  Exports: /home, /soft, /opt
        │  Protocol: NFSv3 & NFSv4
        │  Port: TCP 2049
        │  Network: 192.168.100.0/24
        │
   ┌────┴────────────────────────────┐
   │           │          │           │
hpc-master compute1  compute2    login
(NFS Client) (NFS Client) (NFS Client) (NFS Client)
```

Why storage node as NFS server? The storage node is dedicated solely to storage services. Separating storage from the master node ensures that heavy I/O operations from jobs do not affect cluster management functions running on hpc-master.

Shared directory purposes:

Directory	Purpose
/home	User home directories — accessible from all nodes
/soft	Shared software and custom applications
/opt	Third-party and optional application software

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3. Key Definitions/concepts

NFS Server exports directories and makes them available on the network. It runs the `nfs-server` daemon which handles all client requests.

NFS Client mounts the exported directories from the server. Once mounted, the remote directory appears and behaves like a local directory.

rpcbind is required on both server and client. It maps RPC program numbers to network ports and is a prerequisite for NFS to function.

exportfs is the command used to manage NFS exports. Running `exportfs -arv` re-reads `/etc/exports` and applies all export rules without restarting the NFS service.

/etc/exports is the NFS server configuration file. Each line defines a directory to share, which hosts can access it, and what permissions they have.

/etc/fstab on each client contains persistent mount entries. The `_netdev` option tells the OS to wait for the network to be available before mounting NFS shares during boot.

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4. NFS Server Configuration — storage node

Note: The storage node was installed manually (not via xCAT). All configuration was performed by SSHing directly into the storage node from hpc-master.

4.1 SSH Into Storage Node

```
# From hpc-master
ssh root@storage
```

4.2 Install NFS Server Packages

```
dnf install nfs-utils -y
```

4.3 Create Shared Directories

```
# Create directories
mkdir -p /home
mkdir -p /soft
```

```
mkdir -p /opt

# Set permissions
chmod 755 /home
chmod 755 /soft
chmod 755 /opt

# Verify
ls -ld /home /soft /opt
```

4.4 Configure /etc/exports

```
vi /etc/exports
```

Contents:

```
/home   192.168.100.0/24(rw,sync,no_root_squash,no_subtree_check)
/soft   192.168.100.0/24(rw,sync,no_root_squash,no_subtree_check)
/opt    192.168.100.0/24(rw,sync,no_root_squash,no_subtree_check)
```

Explanation of export options:

Option	Meaning
`rw`	Clients can both read and write to the share
`sync`	Data is written to disk before the server replies to the client — ensures data integrity
`no_root_squash`	The root user on a client has root-level access on the NFS share
`no_subtree_check`	Disables subtree checking — improves reliability and performance
`192.168.100.0/24`	Only hosts on the cluster network can mount these shares

4.5 Start and Enable NFS Services

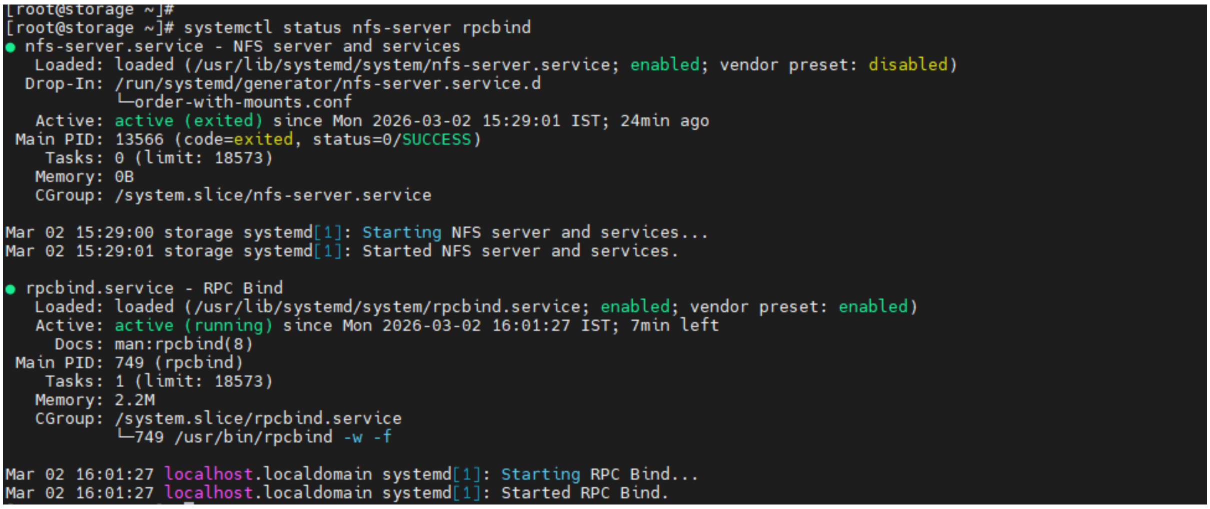
```
# Enable and start NFS server
systemctl enable --now nfs-server

# Enable and start rpcbind
systemctl enable --now rpcbind

# Verify both services
systemctl status nfs-server rpcbind --no-pager
```

Output confirmed:

```
nfs-server.service — Active: active (exited) — Started NFS server and
services
rpcbind.service    — Active: active (running) — Status: Processing requests
```



4.6 Apply Export Configuration

```
# Export all directories defined in /etc/exports
exportfs -arv
```

Output:

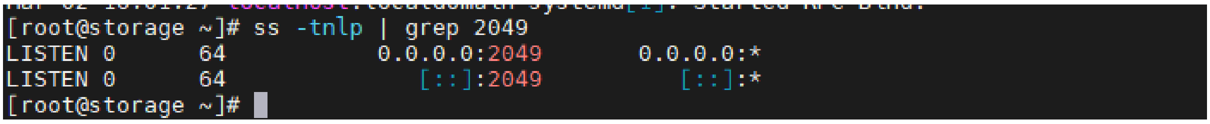
```
exporting 192.168.100.0/24:/opt
exporting 192.168.100.0/24:/soft
exporting 192.168.100.0/24:/home
```

4.7 Verify NFS Server

```
# Check NFS is listening on port 2049
ss -tnlp | grep 2049
```

Output:

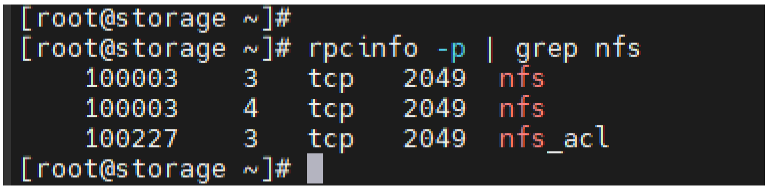
```
LISTEN 0    64    0.0.0.0:2049    0.0.0.0:*
LISTEN 0    64       [::]:2049       [::]:*
```




```
# Verify RPC services are registered
rpcinfo -p | grep nfs
```

Output:

```
100003    3   tcp    2049  nfs
100003    4   tcp    2049  nfs
100227    3   tcp    2049  nfs_acl
```

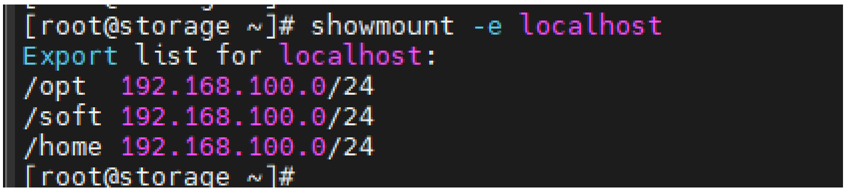


```
[root@storage ~]#
[root@storage ~]# rpcinfo -p | grep nfs
    100003    3   tcp   2049  nfs
    100003    4   tcp   2049  nfs
    100227    3   tcp   2049  nfs_acl
[root@storage ~]#
```

```
# Confirm export list
showmount -e localhost
```

Output:

```
Export list for localhost:
/opt  192.168.100.0/24
/soft 192.168.100.0/24
/home 192.168.100.0/24
```



```
[root@storage ~]# showmount -e localhost
Export list for localhost:
/opt  192.168.100.0/24
/soft 192.168.100.0/24
/home 192.168.100.0/24
[root@storage ~]#
```

NFS Server is fully operational.

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5. NFS Client Configuration

The same steps were applied on all four client nodes — **hpc-master, compute1, compute2, and login** — by SSHing into each node individually from hpc-master.

Note: NFS clients were configured manually via SSH, not using xCAT tools.

5.1 SSH Into Each Client Node

```
ssh root@hpc-master    # already on master, or open new terminal
ssh root@compute1
ssh root@compute2
ssh root@login
```

5.2 Install NFS Client Packages

```
dnf install nfs-utils -y
```

5.3 Create Mount Point Directories

```
mkdir -p /home
mkdir -p /soft
mkdir -p /opt
```

5.4 Test Manual Mount

Before making mounts permanent, test them manually first:

```
mount -t nfs 192.168.100.12:/home /home
mount -t nfs 192.168.100.12:/soft /soft
mount -t nfs 192.168.100.12:/opt  /opt
```

Verify mounts are active:

```
df -h | grep 192.168.100.12
mount | grep nfs
```

Expected output:

```
192.168.100.12:/home   xG   xG   xG   x%   /home
192.168.100.12:/soft   xG   xG   xG   x%   /soft
192.168.100.12:/opt    xG   xG   xG   x%   /opt
```

```
[root@compute2 ~]# df -h
Filesystem               Size  Used Avail Use% Mounted on
devtmpfs                 1.5G     0  1.5G   0% /dev
tmpfs                    1.5G     0  1.5G   0% /dev/shm
tmpfs                    1.5G   17M  1.5G   2% /run
tmpfs                    1.5G     0  1.5G   0% /sys/fs/cgroup
/dev/mapper/xcatvg-root   17G  2.1G   15G  13% /
/dev/nvme0n1p1          1014M  231M  784M  23% /boot
tmpfs                    295M     0  295M   0% /run/user/0
192.168.100.12:/home      56G  2.4G   54G   5% /home
192.168.100.12:/soft      56G  2.4G   54G   5% /soft
192.168.100.12:/opt       56G  2.4G   54G   5% /opt
[root@compute2 ~]#
```

```
192.168.100.12:/opt       56G  2.4G   54G   5% /opt
[root@compute2 ~]# mount | grep nfs
sunrpc on /var/lib/nfs/rpc_pipefs type rpc_pipefs (rw,relatime)
192.168.100.12:/home on /home type nfs4 (rw,relatime,vers=4.2,rsize=524288,wsize=524288,namlen=255,hard,proto=tcp,timeo=600,retrans=2,sec
=sys,clientaddr=192.168.100.14,local_lock=none,addr=192.168.100.12)
192.168.100.12:/soft on /soft type nfs4 (rw,relatime,vers=4.2,rsize=524288,wsize=524288,namlen=255,hard,proto=tcp,timeo=600,retrans=2,sec
=sys,clientaddr=192.168.100.14,local_lock=none,addr=192.168.100.12)
192.168.100.12:/opt on /opt type nfs4 (rw,relatime,vers=4.2,rsize=524288,wsize=524288,namlen=255,hard,proto=tcp,timeo=600,retrans=2,sec=s
ys,clientaddr=192.168.100.14,local_lock=none,addr=192.168.100.12)
[root@compute2 ~]#
```

5.5 Make Mounts Persistent via /etc/fstab

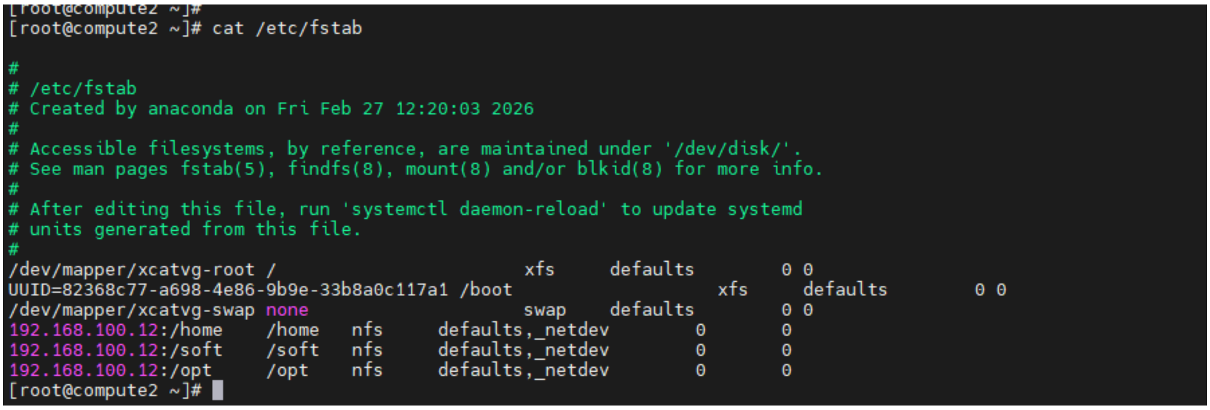
```
echo "192.168.100.12:/home  /home  nfs  defaults,_netdev  0 0" >>
/etc/fstab
echo "192.168.100.12:/soft  /soft  nfs  defaults,_netdev  0 0" >>
/etc/fstab
echo "192.168.100.12:/opt   /opt   nfs  defaults,_netdev  0 0" >>
/etc/fstab
```

Verify fstab entries:

```
Cat /etc/fstab
```

Expected output:

```
#
# /etc/fstab
# Created by anaconda on Fri Feb 27 12:20:03 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more
info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
/dev/mapper/xcatvg-root /                       xfs     defaults        0 0
UUID=82368c77-a698-4e86-9b9e-33b8a0c117a1 /boot                   xfs
defaults        0 0
/dev/mapper/xcatvg-swap none                    swap    defaults        0 0
192.168.100.12:/home    /home   nfs     defaults,_netdev        0       0
192.168.100.12:/soft    /soft   nfs     defaults,_netdev        0       0
192.168.100.12:/opt     /opt    nfs     defaults,_netdev        0       0
```



```
[root@compute2 ~]# cat /etc/fstab

#
# /etc/fstab
# Created by anaconda on Fri Feb 27 12:20:03 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
/dev/mapper/xcatvg-root /                       xfs     defaults        0 0
UUID=82368c77-a698-4e86-9b9e-33b8a0c117a1 /boot                   xfs     defaults        0 0
/dev/mapper/xcatvg-swap none                    swap    defaults        0 0
192.168.100.12:/home    /home   nfs     defaults,_netdev        0       0
192.168.100.12:/soft    /soft   nfs     defaults,_netdev        0       0
192.168.100.12:/opt     /opt    nfs     defaults,_netdev        0       0
[root@compute2 ~]#
```

Explanation of fstab options:

Option	Meaning
`nfs`	Filesystem type is NFS
`defaults`	Use default mount options (rw, auto, exec, etc.)
`_netdev`	Wait for network to be up before mounting — critical for NFS on boot
`0 0`	Do not dump or fsck this filesystem

5.6 Enable rpcbind on Client

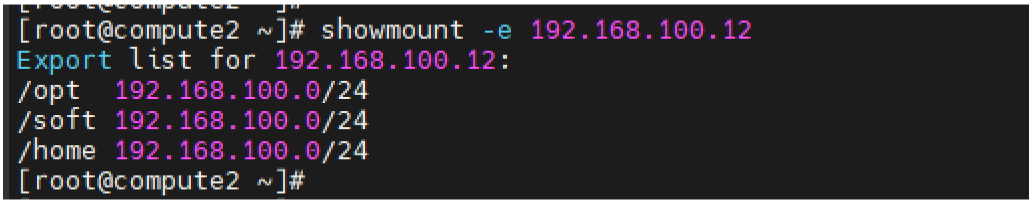
```
systemctl enable --now rpcbind
```

5.7 Verify from Client Side

```
# Check available exports from server
showmount -e 192.168.100.12
```

Expected output:

```
Export list for 192.168.100.12:
/opt  192.168.100.0/24
/soft 192.168.100.0/24
/home 192.168.100.0/24
```



```
[root@compute2 ~]# showmount -e 192.168.100.12
Export list for 192.168.100.12:
/opt  192.168.100.0/24
/soft 192.168.100.0/24
/home 192.168.100.0/24
[root@compute2 ~]#
```

6. Verification — From hpc-master

After configuring all clients, run these verification commands from hpc-master:

```
# Check NFS mounts on all nodes at once
xdsh hpc-master,compute1,compute2,login "df -h | grep 192.168.100.12"
```

```
[root@hpc-master ~]# xdsh hpc-master,compute1,compute2,login "df -h | grep 192.168.100.12"
hpc-master: 192.168.100.12:/home   56G  2.4G   54G   5% /home
hpc-master: 192.168.100.12:/soft   56G  2.4G   54G   5% /soft
hpc-master: 192.168.100.12:/opt    56G  2.4G   54G   5% /opt
[hpc-master]: hpc-master: df: /run/user/0/gvfs: Transport endpoint is not connected
hpc-master: df: /run/user/0/gvfs: Transport endpoint is not connected
compute1: 192.168.100.12:/home      56G  2.4G   54G   5% /home
compute1: 192.168.100.12:/soft      56G  2.4G   54G   5% /soft
compute1: 192.168.100.12:/opt       56G  2.4G   54G   5% /opt
login: 192.168.100.12:/home      56G  2.4G   54G   5% /home
login: 192.168.100.12:/soft      56G  2.4G   54G   5% /soft
login: 192.168.100.12:/opt       56G  2.4G   54G   5% /opt
compute2: 192.168.100.12:/home      56G  2.4G   54G   5% /home
compute2: 192.168.100.12:/soft      56G  2.4G   54G   5% /soft
compute2: 192.168.100.12:/opt       56G  2.4G   54G   5% /opt
[root@hpc-master ~]#
```

```
# Verify mount entries in fstab on all nodes
xdsh hpc-master,compute1,compute2,login "tail -3 /etc/fstab"
```

```
[root@hpc-master ~]# xdsh hpc-master,compute1,compute2,login "tail -3 /etc/fstab"
hpc-master: 192.168.100.12:/home        /home   nfs     defaults,_netdev        0       0
hpc-master: 192.168.100.12:/soft        /soft   nfs     defaults,_netdev        0       0
hpc-master: 192.168.100.12:/opt /opt    nfs     defaults,_netdev        0       0
compute1: 192.168.100.12:/home    /home   nfs     defaults,_netdev        0       0
compute1: 192.168.100.12:/soft    /soft   nfs     defaults,_netdev        0       0
compute1: 192.168.100.12:/opt     /opt    nfs     defaults,_netdev        0       0
compute2: 192.168.100.12:/home    /home   nfs     defaults,_netdev        0       0
compute2: 192.168.100.12:/soft    /soft   nfs     defaults,_netdev        0       0
compute2: 192.168.100.12:/opt     /opt    nfs     defaults,_netdev        0       0
login: 192.168.100.12:/home    /home   nfs     defaults,_netdev        0       0
login: 192.168.100.12:/soft    /soft   nfs     defaults,_netdev        0       0
login: 192.168.100.12:/opt     /opt    nfs     defaults,_netdev        0       0
[root@hpc-master ~]#
```

This confirms all nodes are sharing the same /home, /soft, /opt directories.

---

7. Key Commands

Purpose	Command	Run On
View active exports	exportfs -v	Server
Re-apply exports after changes	exportfs -arv	Server
Show available exports	showmount -e <server-ip>	Client or Server
Check NFS port	ss -tnlp | grep 2049	Server
Check RPC services	rpcinfo -p | grep nfs	Server
Mount NFS share manually	mount -t nfs <ip>:<path> <mountpoint>	Client
Unmount NFS share	umount <mountpoint>	Client
Check mounted shares	df -h | grep <server-ip>	Client
Check fstab entries	tail -3 /etc/fstab	Client
Remount all fstab entries	mount -a	Client
Check NFS server status	systemctl status nfs-server	Server
Restart NFS server	systemctl restart nfs-server	Server